

Data Analytics & Business Intelligence Project

End-to-End Pipeline: Python Data Cleaning & Power BI Interactive Dashboard

STUDENT NAME

Mazen Ayman Shawky

DOMAIN & DATASET

Tech & Electronics Sales Analytics

TECH STACK

Python (Pandas) • Power BI •
DAX

TOTAL REVENUE

\$954.30K

▲ Verified Across Global Markets

TOTAL NET PROFIT

\$320.32K

▲ 33.5% Overall Net Margin

TOTAL UNITS SOLD

2,058 Units

✓ 650 Cleaned Transactions

Executive Summary

This final graduation project demonstrates a complete, production-ready data lifecycle—beginning with raw transaction records, performing automated data sanitization and feature engineering via **Python (Pandas)**, and constructing an executive-grade interactive business intelligence report in **Power BI Desktop**.

The analysis empowers decision-makers with actionable visibility into global commercial performance, identifying top-performing technology products, crucial geographical markets, and dynamic consumer segment behavior.

Final Power BI Executive Dashboard

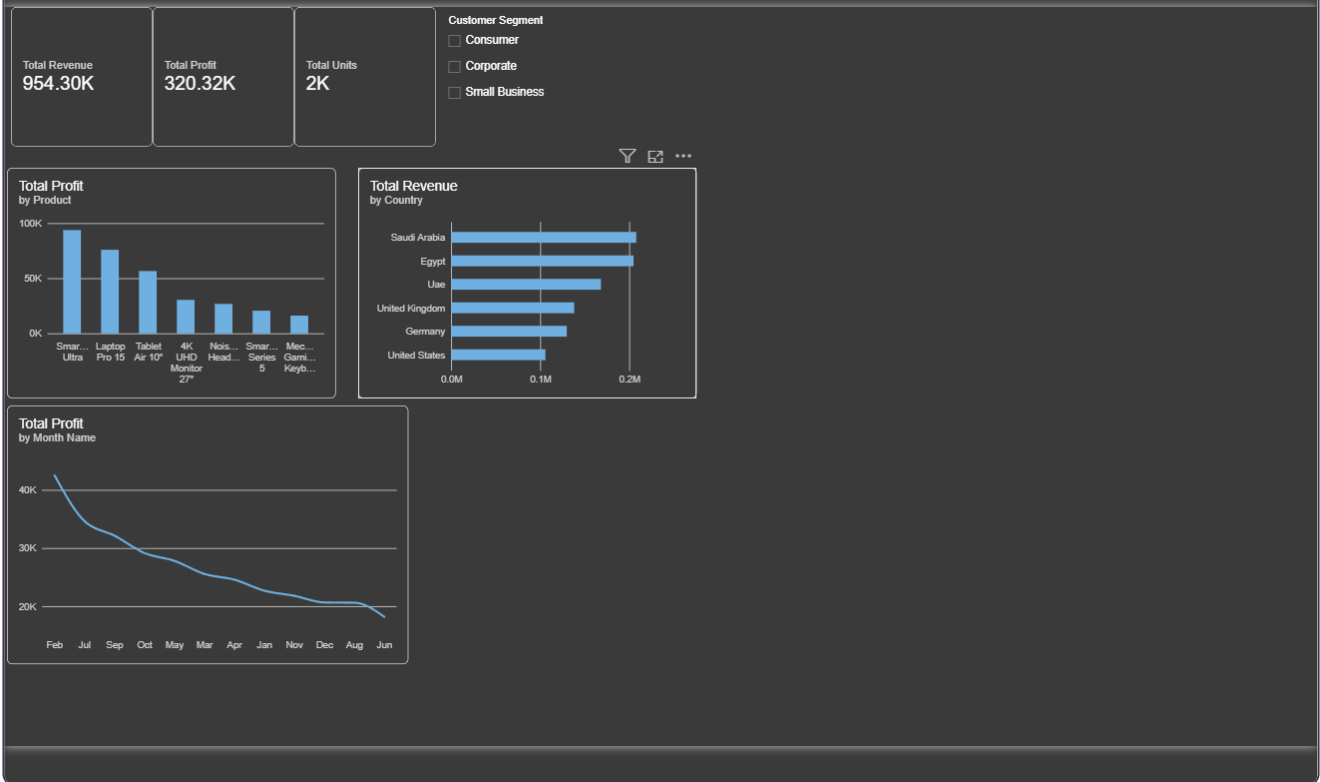


Figure 1.0: Realized Power BI Executive Theme Dashboard developed by Mazen Ayman Shawky.

Pipeline Milestone Achieved: The analytical pipeline guarantees complete data integrity from the command-line script to real-time interactive slicing in Power BI.

1. Data Engineering & Cleaning (Python / Pandas)

The raw data (`tech_sales_raw.csv`) contained inconsistencies, unstructured dates, and required derived performance indicators. An automated Python script (`cleaning.py`) was engineered using the **Pandas** library to enforce standard data types and produce `tech_sales_cleaned.csv`:

- **Missing & Duplicate Handling:** Detected null entries and verified unique transactions against Order ID.
- **Temporal Parsing:** Converted Order Date and Ship Date to native DateTime objects.
- **Operational Feature Engineering:** Calculated Shipping Days = (Ship Date - Order Date).dt.days.
- **Calendar Breakdown:** Extracted month abbreviations (Month Name) to enable monthly trend analysis.
- **Financial Enrichment:** Evaluated the exact transaction profit margin:

```
df['Profit Margin %'] = (df['Profit ($)'] / df['Revenue ($)']) * 100
```

2. Business Logic & DAX Modeling (Power BI)

Explicit DAX measures were constructed in Power BI to ensure dynamic, context-aware aggregation across visuals and slicers:

MEASURE NAME	DAX EXPRESSION	BUSINESS PURPOSE
Total Revenue	SUM(tech_sales_cleaned[Revenue (\$)])	Aggregates gross global revenue (\$954.30K).
Total Profit	SUM(tech_sales_cleaned[Profit (\$)])	Computes net earnings post-cost deduction (\$320.32K).
Total Units	SUM(tech_sales_cleaned[Quantity])	Tracks total volume and logistics load (2,058 Units).

3. Key Insights & Strategic Recommendations

1. **Product Portfolio Drivers:** High-ticket devices represent the overwhelming majority of operating profit. The **Smart Ultra** and **Laptop Pro 15** alone account for more than 50% of cumulative profitability. Conversely, accessories (keyboards, cables) contribute high volume but marginal revenue.

2. **Geographical Strongholds:** The Middle Eastern market—spearheaded by **Saudi Arabia** (~\$210K) and **Egypt** (~\$205K)—is the company's highest yielding territory, closely followed by the UAE and the United Kingdom.
3. **Segment Analysis:** The **Consumer** and **Corporate** segments generate over 75% of sales, while **Small Business** presents an untapped expansion opportunity.
4. **Strategic Recommendations for Management:**
 - Prioritize marketing spend and inventory buffer stock for *Smart Ultra* and *Laptop Pro 15* to mitigate stockout risks during peak cycles.
 - Establish regional fulfillment hubs in Riyadh and Cairo to reduce shipping transit days and capitalize on the strongest market demand.
 - Bundle high-margin hardware with low-cost accessories to lift the average basket value across retail channels.